

PH101: 2023

Special Theory of Relativity

Lecture-4

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Recap

- We have computed the relativistic kinematics like position, time, velocities, under Special Theory of Relativity.
- We will now focus on the dynamical quantities like mass, momentum, energy, acceleration, force etc.

Newton's law and Special Theory of relativity

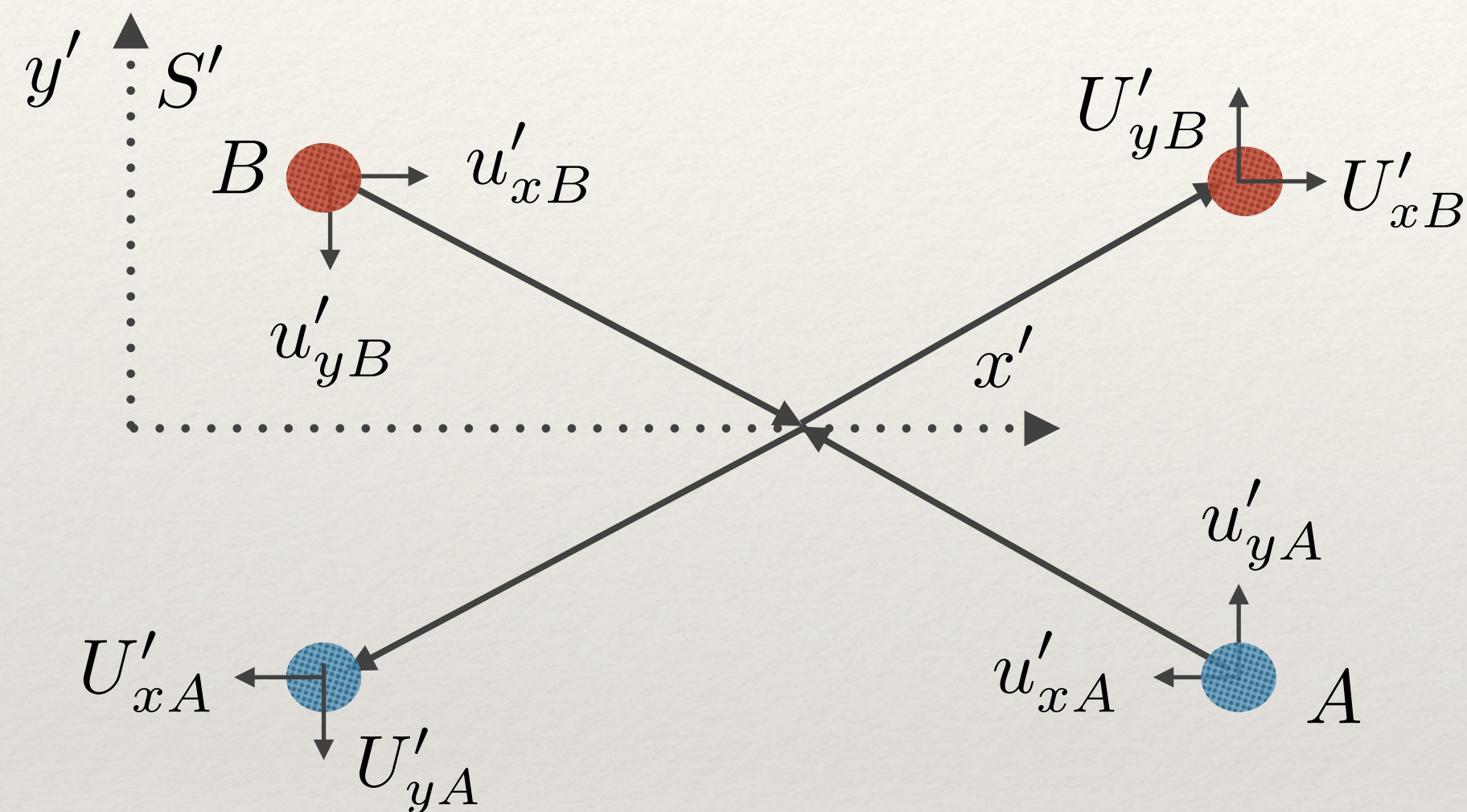
- Newtonian mechanics is inconsistent with Special relativity as its laws are invariant under Galilean transformation, and not invariant under Lorentz transformation.
- In Newtonian mechanics the velocity of a particle can be infinite, where as in STR, the limiting speed is c .
- In classical mechanics, **action-at-a-distance** forces are permitted, while requiring action and reaction forces to be equal. Such equality has no meaning in relativity excepting contact forces, as simultaneity of separated events are relative.



We need to generalize the laws of motion so that it remains invariant under Lorentz transformation and ensure that we obtain the classical limit when $v/c \ll 1$.

Consider: Collision of particles

No action-at-a-distance, the momentum and energy conservation laws are expected to be valid in relativistic approach, so it will help us find the relativistic form of energy, momentum and mass.



Highly symmetrical collision between identical bodies in S' frame:

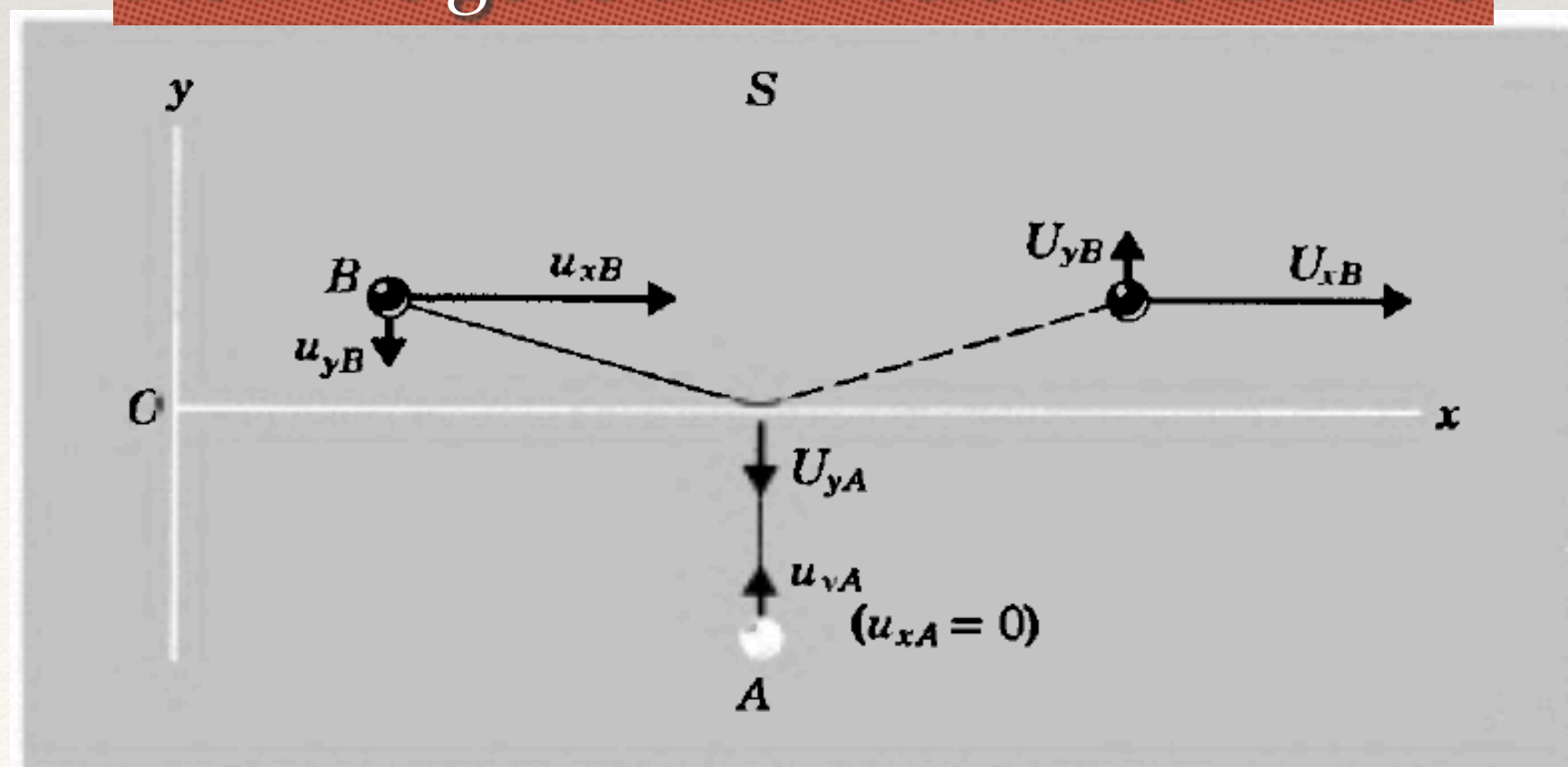
$$\mathbf{u}_{yA}' = -\mathbf{u}_{yB}' \text{ and } \mathbf{u}_{xA}' = -\mathbf{u}_{xB}'$$

The collision is elastic, the final velocities have the same magnitude as the initial velocities, the total momentum after collision remaining zero.

$$\mathbf{u}_{yA}' = -\mathbf{U}_{yA}' = \mathbf{U}_{yB}' = -\mathbf{u}_{yB}' \text{ and } \mathbf{u}_{xA}' = \mathbf{U}_{xA}' = -\mathbf{U}_{xB}' = -\mathbf{u}_{xB}'$$

Sign Flips Before and after collision Remains same

Now we go to a new reference frame S



As seen by observer S , the reference frame S' is moving to the right with a speed v . We deliberately choose $\mathbf{v} = \mathbf{u}_{xB}' = -\mathbf{u}_{xA}'$

Then there is no x-component of motion of A in S frame. $u_{xA} = 0$

The y-component of motion remains unaffected following Galilean relativity,

$$\mathbf{u}_{yA} = \mathbf{u}_{yA}', \mathbf{u}_{yB} = \mathbf{u}_{yB}', \mathbf{u}_{yA} = -\mathbf{U}_{yA} \text{ and } \mathbf{u}_{yB} = -\mathbf{U}_{yB}$$

Elastic Collision: continued

The momentum lost by body A equals that gained by body B

$$2mu_{yA} = 2mu_{yB} \quad u_{yA} = u_{yB}.$$

The question is whether these results are consistent with Lorentz transformation.

Following the velocity addition formulae,

For body B:

$$u_{yB}' = \frac{u_{yB} \sqrt{1 - \beta^2}}{1 - u_{xB}v/c^2}$$

$$\text{Recall, } u_y' = \frac{u_y}{\gamma(1 - u_x v/c^2)}$$

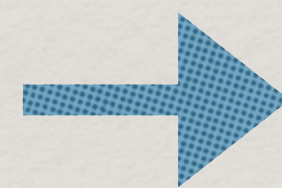
For body A:

$$u_{yA}' = u_{yA} \sqrt{1 - \beta^2}$$

$$u_{xA} = 0$$

Assuming,

$$u_{yB}' = u_{yA}'$$



$$u_{yA} = u_{yB} \frac{1}{1 - u_{xB}v/c^2}$$

This is in contrast to the Newtonian result,

$$u_{yA} = u_{yB}.$$

Therefore if we compute momentum by $p=mv$, then momentum is conserved in one frame, but not in the other.

Therefore, the result is in contrast with the basic postulates of Special relativity that laws of Physics are same in different frames.

We will now show that classical definition of momentum and/or mass need to be redefined, so that we can restore momentum conservation.

Relativistic mass

Momentum conservation:

$$2m_A u_{yA} = 2m_B u_{yB}$$

$$m_B = m_A \frac{u_{yA}}{u_{yB}} = \frac{m_A}{1 - u_{xB}v/c^2}$$

using

$$u_{yA} = u_{yB} \frac{1}{1 - u_{xB}v/c^2}$$

The relativistic masses are not equal, if the relativistic momentum conservation law has the same form.

Using relativistic velocity addition,

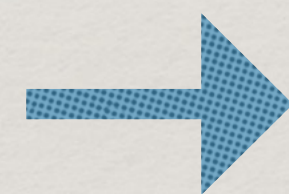
$$u_{xB}' (= v) = \frac{u_{xB} - v}{1 - u_{xB}v/c^2}.$$

Solving:

$$v = \frac{c^2}{u_{xB}} (1 - \sqrt{1 - (u_{xB}/c)^2}).$$

Substituting,

$$m_B = \frac{m_A}{\sqrt{1 - (u_{xB}/c)^2}}.$$



$$m = \frac{m_0}{\sqrt{1 - u^2/c^2}}$$

When $u/c \rightarrow 0$, $m = m_0$

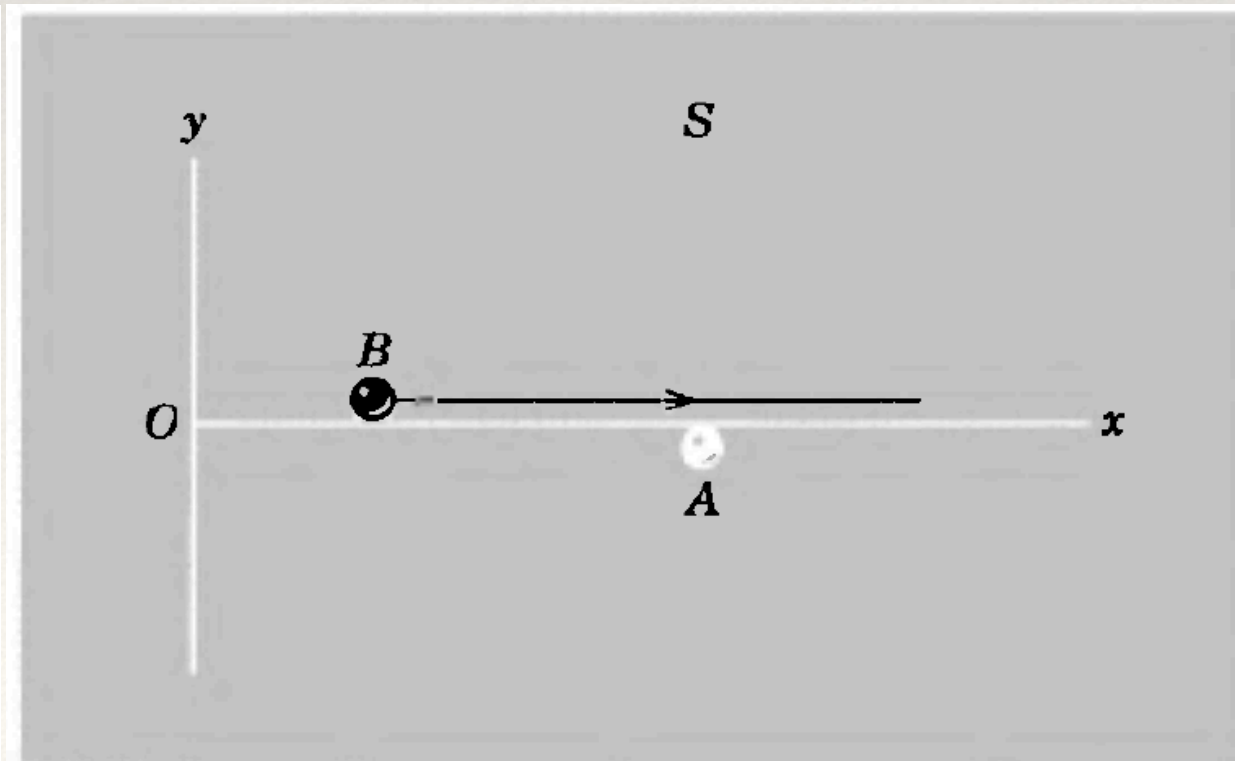
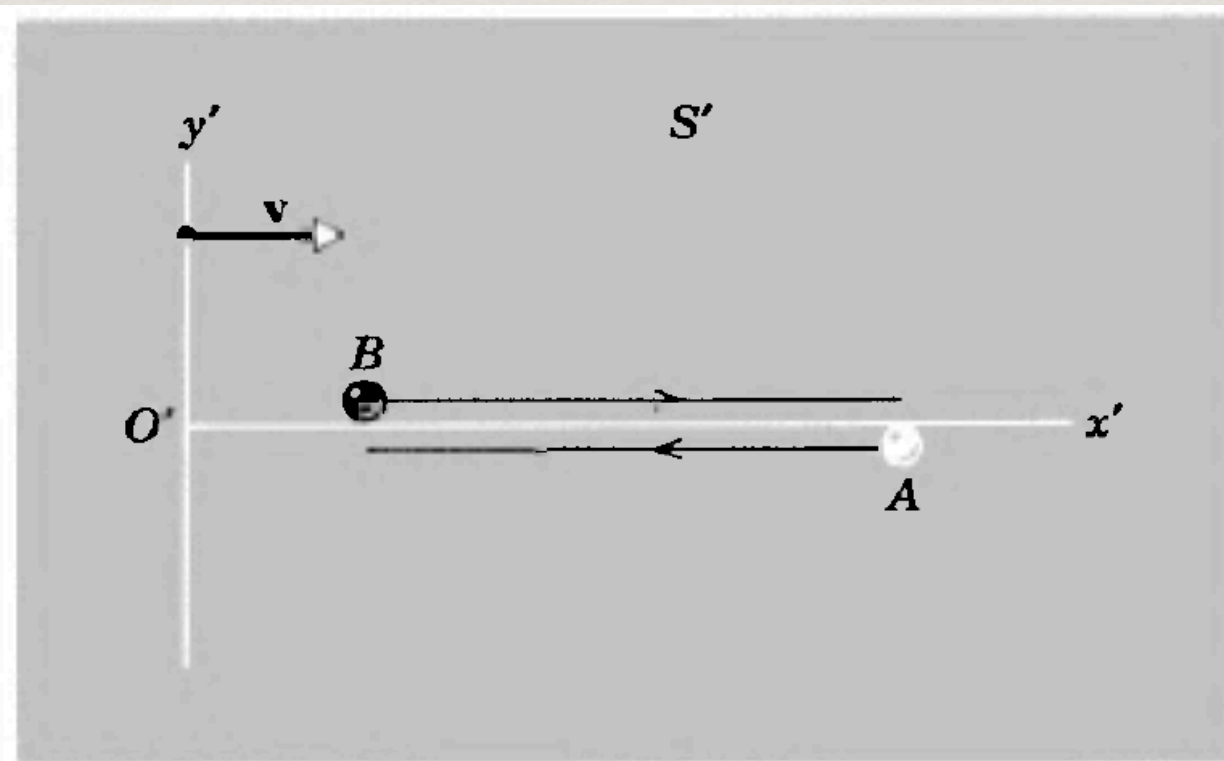
Mass is scalar, so any direction of velocity of the particle provide the same answer.

Let's call the rest mass to be m_0

$$m_A = m_0$$

B has the same mass as A at rest.

Consider the y-component of velocity very very small, tends to zero.



Relativistic Momentum

One can show that using the form of the relativistic mass, if the momentum is defined as $\mathbf{p} = m\mathbf{v}$, then it is conserved in all the inertial frames.

$$\mathbf{p} = \frac{m_0 \mathbf{u}}{\sqrt{1 - u^2/c^2}} \quad \rightarrow \quad p_x = \frac{m_0 u_x}{\sqrt{1 - u^2/c^2}} \quad p_y = \frac{m_0 u_y}{\sqrt{1 - u^2/c^2}} \quad p_z = \frac{m_0 u_z}{\sqrt{1 - u^2/c^2}}$$

$$\vec{p} = \gamma m_0 \vec{u} \quad \gamma = \frac{1}{\sqrt{1 - u^2/c^2}} = \frac{1}{\sqrt{1 - \beta^2}} \quad \beta = u/c$$

Therefore, $\mathbf{F} = \frac{d}{dt}(\mathbf{p}) = \frac{d}{dt} \left(\frac{m_0 \mathbf{u}}{\sqrt{1 - u^2/c^2}} \right) \neq (m_0 / \sqrt{1 - u^2/c^2}) (d\mathbf{u}/dt), \quad \vec{F} \neq m\vec{a}$

For example,

$$q(\mathbf{E} + \mathbf{u} \times \mathbf{B}) = \frac{d}{dt} \left(\frac{m_0 \mathbf{u}}{\sqrt{1 - u^2/c^2}} \right)$$

Kinetic Energy

In **Newtonian mechanics** the **kinetic energy**, K , of a particle is equal to the work done by an external force in increasing the speed of the particle from zero to some value u . That is,

$$K = \int_{u=0}^{u=u} \mathbf{F} \cdot d\mathbf{l} = \int_{u=0}^{u=u} F dx = \int m_0 \left(\frac{du}{dt} \right) dx = \int m_0 du \frac{dx}{dt} = m_0 \int_0^u u du = \frac{1}{2} m_0 u^2.$$

m_0 is independent of the speed of the particle. The force above is defined as

$$F = m_0 a = m_0 (du/dt)$$

Relativistically,

$$\begin{aligned} K &= \int_{u=0}^{u=u} F dx = \int \frac{d}{dt} (mu) dx = \int d(mu) \frac{dx}{dt} \\ &= \int (m du + u dm) u = \int_{u=0}^{u=u} (mu du + u^2 dm) \end{aligned}$$

Differentiating and rearranging, we get

$$\begin{aligned} 2mc^2 dm - m^2 2u du - u^2 2m dm &= 0 \\ mu du + u^2 dm &= c^2 dm \end{aligned}$$

$$K = \int_{u=0}^{u=u} c^2 dm = c^2 \int_{m=m_0}^{m=m} dm = mc^2 - m_0 c^2$$

$$K = m_0 c^2 \left[\frac{1}{\sqrt{1 - u^2/c^2}} - 1 \right]$$

$$m = m_0 / \sqrt{1 - u^2/c^2}$$

$$m^2 c^2 - m^2 u^2 = m_0^2 c^2$$

$$E = mc^2$$



$$E = m_0 c^2 + K$$

Energy Momentum relations in STR

Dynamical quantities in STR

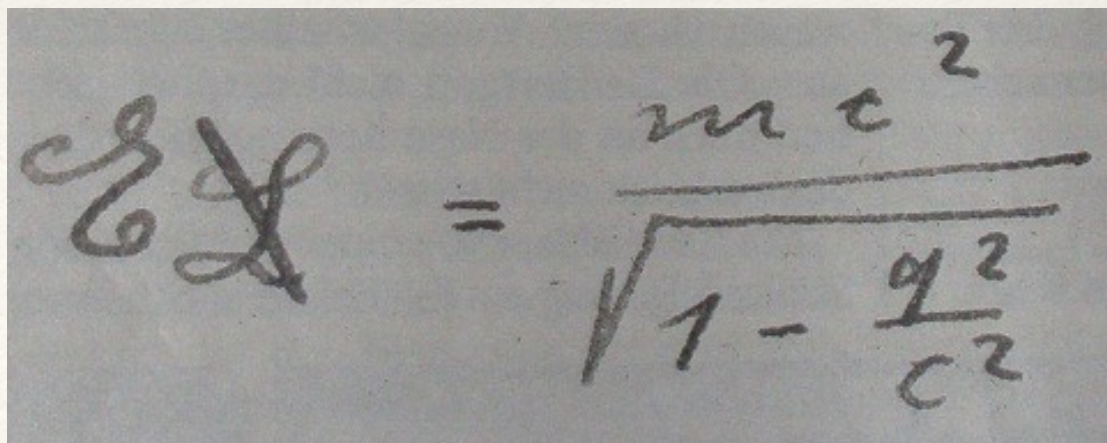
$$\vec{p} = m\vec{u} = m_0\gamma\vec{u}$$

$$E = mc^2 = m_0\gamma c^2$$

$$K = mc^2 - m_0c^2 = m_0c^2(\gamma - 1)$$

$$E^2 = p^2c^2 + m_0^2c^4$$

Equivalence of mass and energy. For example, electron mass is 0.5 MeV.



A photograph of a handwritten equation on a piece of paper. The equation is $E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}}$. The handwriting is in cursive, and the paper appears to be a scan of a physical document.

Einstein's own
handwriting
(1912)

$$p^2 = \frac{m_0^2 u^2}{1 - u^2/c^2} = \frac{m_0^2 u^2 c^2}{c^2 - u^2}$$

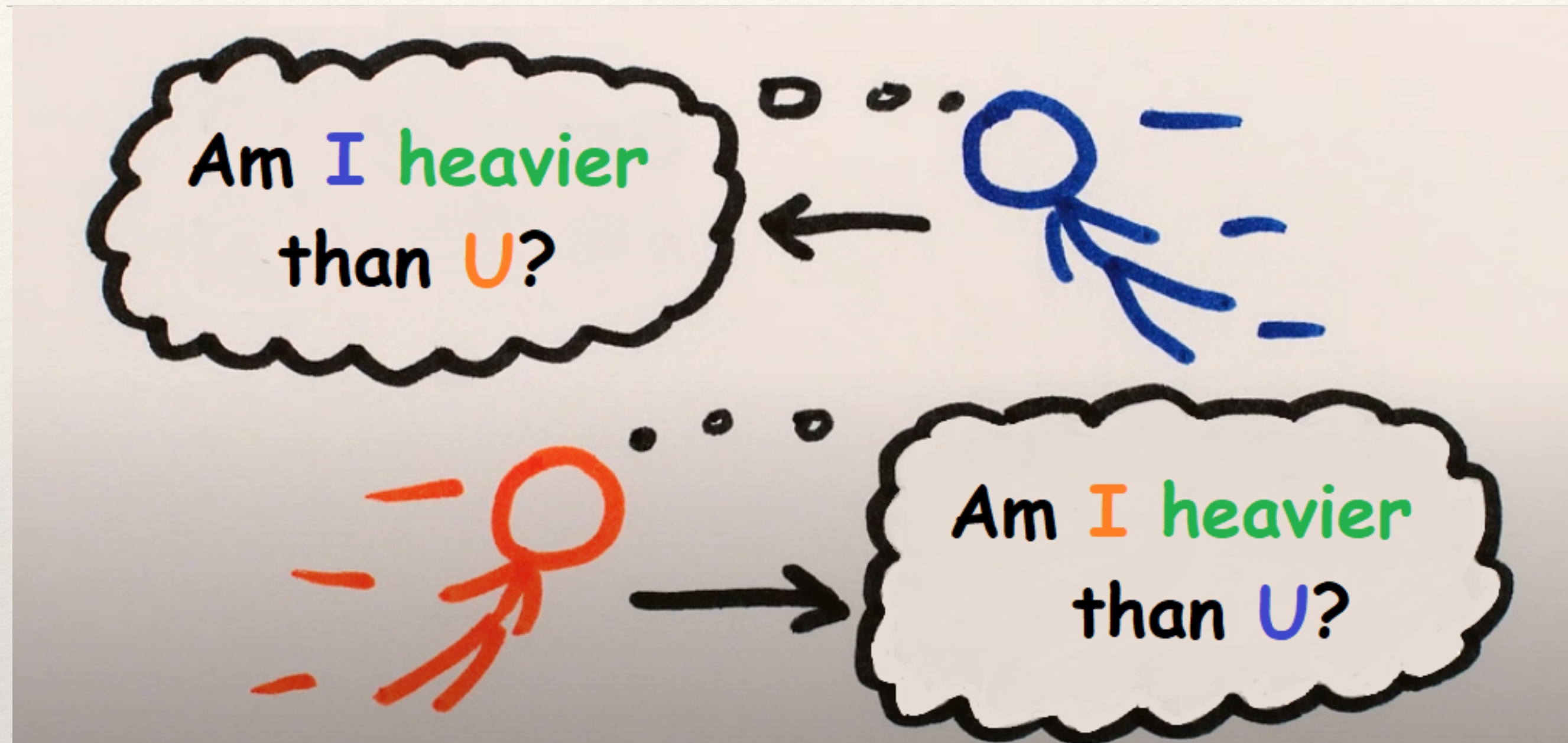
$$\frac{u^2}{c^2} = \frac{p^2}{p^2 + m_0^2 c^2}$$

$$\gamma = \frac{1}{\sqrt{(1 - u^2/c^2)}} = \sqrt{1 + \frac{p^2}{m_0^2 c^2}}$$

$$E^2 = m_0^2 \gamma^2 c^4$$

$$E^2 = p^2 c^2 + m_0^2 c^4$$

Relativistic mass, energy and momentum



In a letter to Lincoln Barnett, Einstein said,

"It is not good to introduce the concept of the mass $M = m / \sqrt{1 - \frac{v^2}{c^2}}$ of a moving body for which no clear definition can be given. It is better to introduce no other mass concept than the 'rest mass' m . Instead of introducing M , it is better to mention the expression for the momentum and energy of a body in motion."

There is a debate regarding the idea of the mass changing from one inertial frame to the other via the concept of relativistic mass, as it is an inbuilt characteristics / composition of an object.

However, the definition of relativistic momenta and energy perfectly captures the postulate of STR, as both energy and momentum conservation is applicable to any inertial frames, if conserved in one.

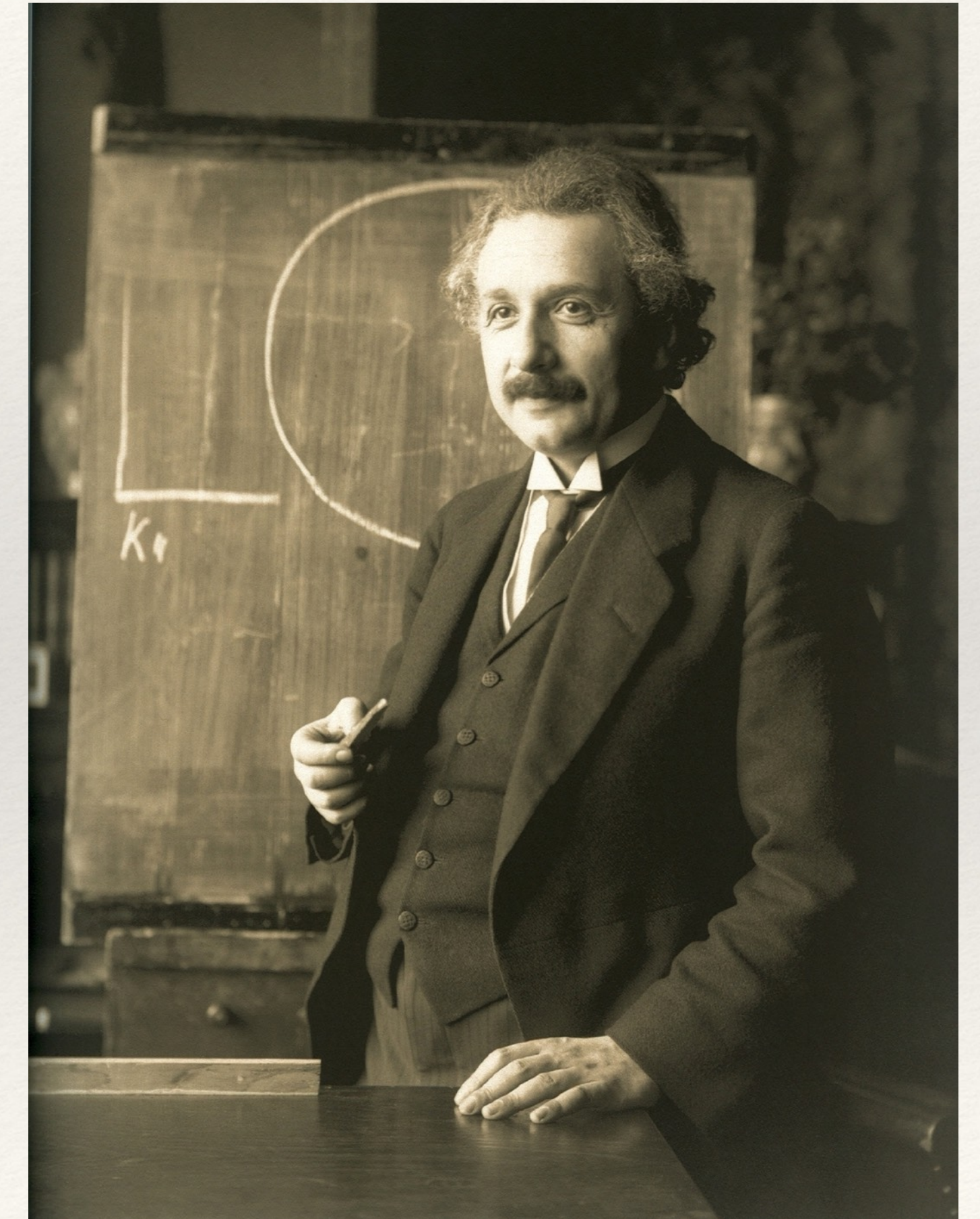
Classical limit of relativistic momentum

If $u/c \ll 1$, the relativistic expression for K reduces to the classical result,

$$K = mc^2 - m_0c^2 = m_0c^2(\gamma - 1)$$

$$\begin{aligned} K &= m_0c^2 \left[\left(1 - \frac{u^2}{c^2} \right)^{-1/2} - 1 \right] \\ &= m_0c^2 \left[\left(1 - \frac{u^2}{c^2} \right)^{-1/2} - 1 \right] \\ &= m_0c^2 \left[1 + \frac{1}{2} \left(\frac{u}{c} \right)^2 + \frac{3}{8} \left(\frac{u}{c} \right)^4 + \dots - 1 \right] \\ &= \frac{1}{2}m_0u^2 \end{aligned}$$

Note that the relativistic formula for kinetic energy is not $\frac{1}{2}mu^2$. We cannot simply substitute the relativistic mass for the rest mass in generalizing a classical formula to a relativistic one.



Example

Example: Compute the momentum of a 10 Mev electron, both classically and relativistically.

Classically,

$$\begin{aligned} p &= \sqrt{2m_0K} \\ &= \sqrt{2 \times 9.1 \times 10^{-31} \text{ kg} \times 10 \text{ Mev} \times 1.6 \times 10^{-13} \text{ joule/Mev}} \\ &= 17 \times 10^{-22} \text{ kg m/sec} \end{aligned}$$

Relativistically,

$$\begin{aligned} p &= \frac{1}{c} \sqrt{(K + m_0c^2)^2 - (m_0c^2)^2}, \quad m_0c^2 = 0.51 \text{ Mev.} \\ &= \frac{1}{3 \times 10^8} \sqrt{(10 + 0.51)^2 - (0.51)^2} \frac{\text{Mev-sec}}{\text{meter}} (1.16 \times 10^{-13} \text{ joule/Mev}) \\ &= 5.6 \times 10^{-21} \text{ kg - m/sec} \end{aligned}$$

Massless particle

Massless particles are particles which possess momentum and energy but no rest mass. If we take $m_0 = 0$ in the relation

$$E^2 = (pc)^2 + (m_0c^2)^2,$$

in the limit $m_0 \rightarrow 0$. This is only possible if $u \rightarrow c$ as $m_0 \rightarrow 0$.

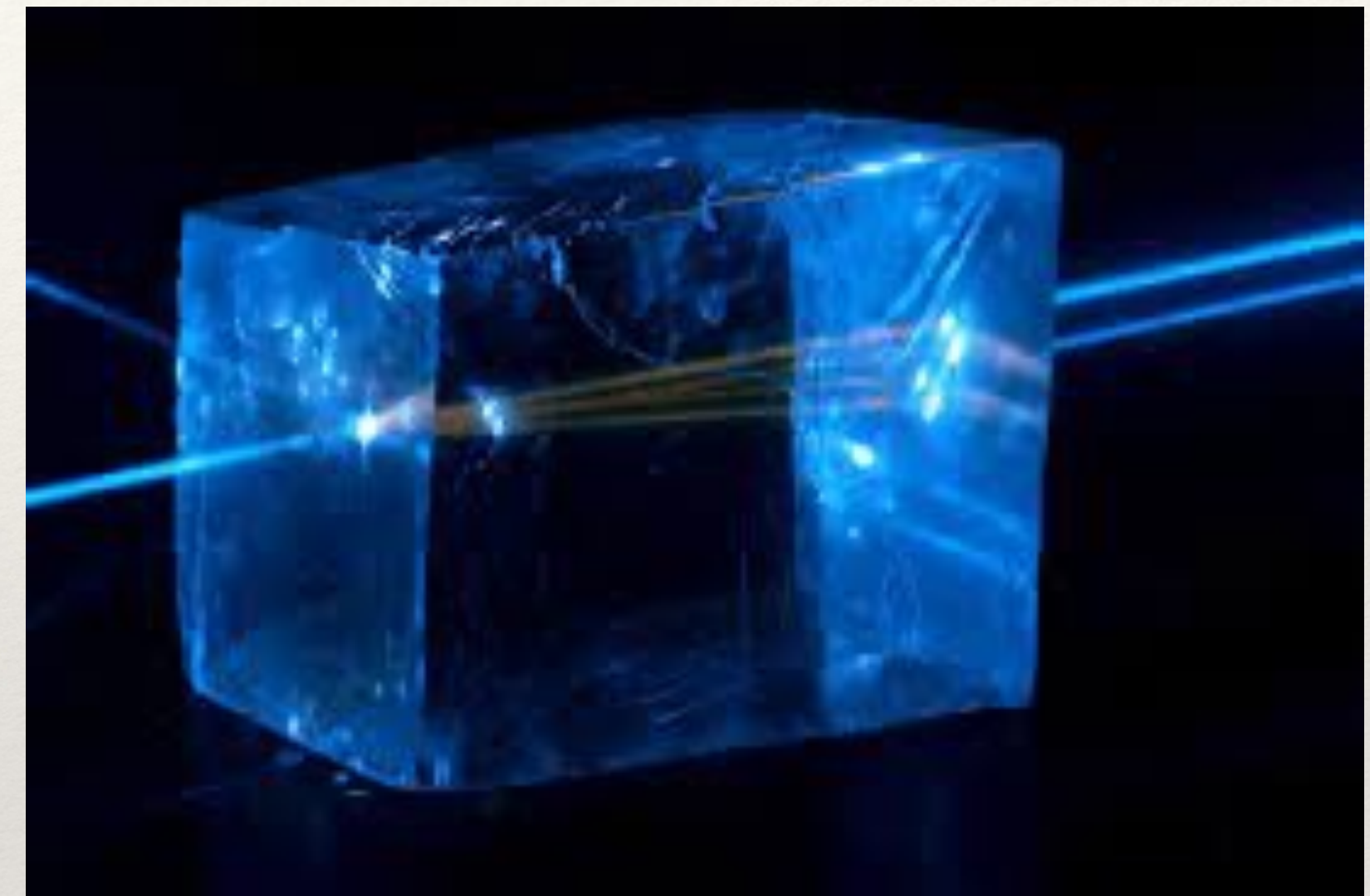
$$E = pc$$

To have nonzero momentum we must have a finite value for

$$p = \frac{m_0 u}{\sqrt{1 - \frac{u^2}{c^2}}}$$

in the limit $m_0 \rightarrow 0$. This is only possible if $u \rightarrow c$ as $m_0 \rightarrow 0$.

Massless particles must travel at the speed of light.



Photon the quantum particle responsible for electromagnetic interaction, is massless and moves with the velocity of light.

Summary

- We defined relativistic mass, which keeps momentum conserved in all frames

$$m = \frac{m_0}{\sqrt{1 - u^2/c^2}}$$

- We defined relativistic momentum, following relativistic mass as:

$$\mathbf{p} = \frac{m_0 \mathbf{u}}{\sqrt{1 - u^2/c^2}} = \vec{p} = \gamma m_0 \vec{u}$$

- Mass-energy equivalence relation, which in turn provides a relativistic expression for kinetic energy:

$$E = mc^2$$

$$E = m_0 c^2 + K$$

$$E^2 = p^2 c^2 + m_0^2 c^4$$